Week 8: Database Relationship Analysis using Joins

Scenario:

The online commerce platform maintains customer accounts, inventory products, purchasing orders, and payment records across normalized relational entities. The business intelligence team seeks to correlate these interdependent tables to gain clear visibility into client shopping patterns, order fulfillment timelines, and payment clearance status.

Implement INNER JOIN

Concept:

An INNER JOIN returns only records that possess corresponding matches in both participating tables.

Tasks:

1. Display customer details along with their orders.

```
SQL> SELECT CONCAT(C.FIRST_NAME, C.LAST_NAME) AS CUSTOMER_NAME,
  2  O.ORDER_REF,
  3  O.ORDER_TOTAL
  4  FROM CUSTOMERS C
  5  INNER JOIN ORDERS O
  6  ON C.CUST_ID = O.CUST_ID;

CUSTOMER_NAME                  ORDER_REF ORDER_TOTAL
------------------------------ --------- -----------
ArjunKumar                     ORD0001         55850
PriyaSharma                    ORD0002          3350
RahulMenon                     ORD0003         28000
AnanyaIyer                     ORD0004          8300
VikramReddy                    ORD0005         22000
MeeraKrishnan                  ORD0006          7000
SureshNair                     ORD0007          6500
KavyaRajan                     ORD0008          5700
KarthikSubramanian             ORD0009         18500
DivyaPillai                    ORD0010          5200
AdityaVerma                    ORD0011          7000

CUSTOMER_NAME                  ORDER_REF ORDER_TOTAL
------------------------------ --------- -----------
SnehaPatel                     ORD0012          2450
LakshmiVenkatesan              ORD0014         28000
NithinMurugan                  ORD0015          8350

14 rows selected.
```

2. Display order details with payment information.

```
SQL> SELECT O.ORDER_REF,
  2  O.ORDER_TOTAL,
  3  O.ORDER_DATE,
  4  O.CUST_ID,
  5  P.TXN_DATE,
  6  P.PAY_METHOD,
  7  P.TXN_STATUS
  8  FROM ORDER_PAYMENTS P JOIN ORDERS O
  9  ON P.ORDER_REF = O.ORDER_REF;

ORDER_REF ORDER_TOTAL ORDER_DATE CUST_ID  TXN_DATE  PAY_METHOD      TXN_STATUS
--------- ----------- ---------- -------- --------- --------------- ----------
ORD0001         55850 10-JAN-26  CUST0001 10-JAN-26 Credit Card     Completed
ORD0002          3350 25-JAN-26  CUST0002 25-JAN-26 UPI             Completed
ORD0004          8300 18-FEB-26  CUST0004 18-FEB-26 Debit Card      Completed
ORD0006          7000 15-MAR-26  CUST0006 15-MAR-26 Net Banking     Completed
ORD0008          5700 20-APR-26  CUST0008 20-APR-26 COD             Completed
ORD0010          5200 18-MAY-26  CUST0010 18-MAY-26 UPI             Completed
ORD0012          2450 20-JUN-26  CUST0012 20-JUN-26 Credit Card     Completed
ORD0014         28000 15-AUG-26  CUST0014 15-AUG-26 Debit Card      Completed
ORD0014         28000 15-AUG-26  CUST0014 14-AUG-26 Debit Card      Failed

9 rows selected.
```

3. Retrieve products purchased by customers.

```
SQL> SELECT
  2       C.CUST_ID,
  3       C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4       O.ORDER_REF,
  5       P.PROD_ID,
  6       P.ITEM_NAME,
  7       OD.ITEM_QTY,
  8       OD.UNIT_PRICE,
  9       OD.ITEM_QTY * OD.UNIT_PRICE AS TOTAL_AMOUNT
 10  FROM CUSTOMERS C
 11  JOIN ORDERS O
 12    ON C.CUST_ID = O.CUST_ID
 13  JOIN ORDER_ITEMS OD
 14    ON O.ORDER_REF = OD.ORDER_REF
 15  JOIN PRODUCTS P
 16    ON OD.PROD_ID = P.PROD_ID
 17  ORDER BY C.CUST_ID, O.ORDER_DATE;

CUST_ID  CUSTOMER_NAME                   ORDER_REF PROD_ID
---------------------------------------- --------- -------
ITEM_NAME                                ITEM_QTY UNIT_PRICE
---------------------------------------- -------- ----------
TOTAL_AMOUNT
------------
CUST0001 Arjun Kumar                     ORD0001   P000001
Laptop                                          1      55000
       55000

CUST0001 Arjun Kumar                     ORD0001   P000002
Wireless Mouse                                  1        850
         850

CUST_ID  CUSTOMER_NAME                   ORDER_REF PROD_ID
---------------------------------------- --------- -------
ITEM_NAME                                ITEM_QTY UNIT_PRICE
---------------------------------------- -------- ----------
TOTAL_AMOUNT
------------
CUST0002 Priya Sharma                    ORD0002   P000006
USB Cable                                       1        450
         450

CUST0002 Priya Sharma                    ORD0002   P000004
Bluetooth Speaker                               1       3200
        3200

CUST_ID  CUSTOMER_NAME                   ORDER_REF PROD_ID
---------------------------------------- --------- -------
ITEM_NAME                                ITEM_QTY UNIT_PRICE
---------------------------------------- -------- ----------
TOTAL_AMOUNT
------------
CUST0003 Rahul Menon                     ORD0003   P000005
Smartphone                                      6      28000
      168000

CUST0004 Ananya Iyer                     ORD0004   P000006
```

```
ITEM_NAME                           ITEM_QTY UNIT_PRICE
----------------------------------- -------- ----------
TOTAL_AMOUNT
------------
USB Cable                                  4        450
        1800

CUST0004 Ananya Iyer                ORD0004   P000008
Smart Watch                                1       6500
        6500

CUST_ID  CUSTOMER_NAME              ORDER_REF PROD_ID
--------------------------------------- --------- -------
ITEM_NAME                           ITEM_QTY UNIT_PRICE
----------------------------------- -------- ----------
TOTAL_AMOUNT
------------
CUST0005 Vikram Reddy               ORD0005   P000010
Tablet                                     1      22000
       22000

CUST0006 Meera Krishnan             ORD0006   P000015
Wireless Earbuds                           1       3500
        3500

CUST_ID  CUSTOMER_NAME              ORDER_REF PROD_ID
--------------------------------------- --------- -------
ITEM_NAME                           ITEM_QTY UNIT_PRICE
----------------------------------- -------- ----------
TOTAL_AMOUNT
------------
CUST0006 Meera Krishnan             ORD0006   P000009
Headphones                                 1       4200
        4200

CUST0007 Suresh Nair                ORD0007   P000008
Smart Watch                                1       6500
        6500

CUST_ID  CUSTOMER_NAME              ORDER_REF PROD_ID
--------------------------------------- --------- -------
ITEM_NAME                           ITEM_QTY UNIT_PRICE
----------------------------------- -------- ----------
TOTAL_AMOUNT
------------
CUST0008 Kavya Rajan                ORD0008   P000012
External Hard Disk                         1       5200
        5200

CUST0008 Kavya Rajan                ORD0008   P000006
USB Cable                                  1        450
         450

CUST0009 Karthik Subramanian        ORD0009   P000013
Gaming Monitor                             1      18500
       18500

CUST_ID  CUSTOMER_NAME              ORDER_REF PROD_ID
--------------------------------------- --------- -------
ITEM_NAME                           ITEM_QTY UNIT_PRICE
----------------------------------- -------- ----------
TOTAL_AMOUNT
------------
CUST0010 Divya Pillai               ORD0010   P000012
External Hard Disk                         1       5200
        5200

CUST0011 Aditya Verma               ORD0011   P000015
Wireless Earbuds                           1       3500
        3500
```

```
CUST_ID  CUSTOMER_NAME                  ORDER_REF PROD_ID
---------------------------------------- --------- -------
ITEM_NAME                               ITEM_QTY UNIT_PRICE
---------------------------------------- -------- ----------
TOTAL_AMOUNT
------------
CUST0011 Aditya Verma                   ORD0011   P000009
Headphones                                     1       4200
        4200

CUST0012 Sneha Patel                    ORD0012   P000014
Laptop Stand                                   1       1500
        1500

CUST_ID  CUSTOMER_NAME                  ORDER_REF PROD_ID
---------------------------------------- --------- -------
ITEM_NAME                               ITEM_QTY UNIT_PRICE
---------------------------------------- -------- ----------
TOTAL_AMOUNT
------------
CUST0012 Sneha Patel                    ORD0012   P000003
Mechanical Keyboard                            1       2500
        2500

CUST0015 Nithin Murugan                 ORD0015   P000015
Wireless Earbuds                               1       3500
        3500

20 rows selected.
```

Implement LEFT JOIN

Tasks:

1. Display all customers including customers who have not placed orders.

```
SQL> SELECT
  2       C.CUST_ID,
  3       C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4       C.EMAIL_ADDR,
  5       O.ORDER_REF,
  6       O.ORDER_DATE,
  7       O.ORDER_TOTAL,
  8       O.ORDER_STATUS
  9  FROM CUSTOMERS C
 10  LEFT JOIN ORDERS O
 11    ON C.CUST_ID = O.CUST_ID
 12  ORDER BY C.CUST_ID;

CUST_ID  CUSTOMER_NAME
--------------------------------------------------------------------------------
EMAIL_ADDR                     ORDER_REF ORDER_DAT ORDER_TOTAL ORDER_STATUS
------------------------------ --------- --------- ----------- ------------
CUST0001 Arjun Kumar
arjun.kumar@gmail.com          ORD0001   10-JAN-26       55850 Delivered

CUST0002 Priya Sharma
priya.sharma@gmail.com         ORD0002   25-JAN-26        3350 Delivered

CUST0003 Rahul Menon
rahul.menon@gmail.com          ORD0003   05-FEB-26       28000 Delivered

CUST_ID  CUSTOMER_NAME
--------------------------------------------------------------------------------
EMAIL_ADDR                     ORDER_REF ORDER_DAT ORDER_TOTAL ORDER_STATUS
------------------------------ --------- --------- ----------- ------------
CUST0004 Ananya Iyer
ananya.iyer@gmail.com          ORD0004   18-FEB-26        8300 Delivered

CUST0005 Vikram Reddy
vikram.reddy@gmail.com         ORD0005   02-MAR-26       22000 Processing

CUST0006 Meera Krishnan
meera.krishnan@gmail.com       ORD0006   15-MAR-26        7000 Delivered

CUST_ID  CUSTOMER_NAME
--------------------------------------------------------------------------------
EMAIL_ADDR                     ORDER_REF ORDER_DAT ORDER_TOTAL ORDER_STATUS
------------------------------ --------- --------- ----------- ------------
CUST0007 Suresh Nair
suresh.nair@gmail.com          ORD0007   01-APR-26        6500 Shipped

CUST0008 Kavya Rajan
kavya.rajan@gmail.com          ORD0008   20-APR-26        5700 Delivered

CUST0009 Karthik Subramanian
karthik.subramanian@gmail.com  ORD0009   05-MAY-26       18500 Processing

CUST_ID  CUSTOMER_NAME
--------------------------------------------------------------------------------
EMAIL_ADDR                     ORDER_REF ORDER_DAT ORDER_TOTAL ORDER_STATUS
------------------------------ --------- --------- ----------- ------------
CUST0010 Divya Pillai
divya.pillai@gmail.com         ORD0010   18-MAY-26        5200 Delivered

CUST0011 Aditya Verma
aditya.verma@gmail.com         ORD0011   03-JUN-26        7000 Shipped

CUST0012 Sneha Patel
sneha.patel@gmail.com          ORD0012   20-JUN-26        2450 Delivered

CUST_ID  CUSTOMER_NAME
--------------------------------------------------------------------------------
EMAIL_ADDR                     ORDER_REF ORDER_DAT ORDER_TOTAL ORDER_STATUS
------------------------------ --------- --------- ----------- ------------
CUST0013 Rohan Bose
rohan.bose@gmail.com

CUST0014 Lakshmi Venkatesan
lakshmi.venkatesan@gmail.com   ORD0014   15-AUG-26       28000 Delivered

CUST0015 Nithin Murugan
nithin.murugan@gmail.com       ORD0015   25-SEP-26        8350 Processing

15 rows selected.
```

2. Find customers without any purchases.

```
SQL> SELECT
  2       C.CUST_ID,
  3       C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4       C.EMAIL_ADDR,
  5       C.CONTACT_NO
  6  FROM CUSTOMERS C
  7  LEFT JOIN ORDERS O
  8    ON C.CUST_ID = O.CUST_ID
  9  WHERE O.ORDER_REF IS NULL
 10  ORDER BY C.CUST_ID;

CUST_ID  CUSTOMER_NAME        EMAIL_ADDR                    CONTACT_NO
-------- -------------------- ----------------------------- ----------
CUST0013 Rohan Bose           rohan.bose@gmail.com          9654321098
```

3. Display all products including products with no sales.

```
SQL> SELECT
  2       P.PROD_ID,
  3       P.ITEM_NAME,
  4       P.UNIT_PRICE
  5  FROM PRODUCTS P
  6  LEFT JOIN ORDER_ITEMS OD
  7    ON P.PROD_ID = OD.PROD_ID
  8  WHERE OD.PROD_ID IS NULL
  9  ORDER BY P.PROD_ID;

PROD_ID ITEM_NAME                                          UNIT_PRICE
------- -------------------------------------------------- ----------
P000007 Power Bank                                               1800
P000011 Web Camera                                               2800
```

Implement RIGHT JOIN

Concept:

A RIGHT JOIN preserves all records from the right-hand table, pairing them with matching rows from the left-hand table or NULLs.

Tasks:

1. Display all orders with customer information.

```
SQL> SELECT
  2       O.ORDER_REF,
  3       O.ORDER_DATE,
  4       O.ORDER_TOTAL,
  5       O.ORDER_STATUS,
  6       C.CUST_ID,
  7       C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  8       C.EMAIL_ADDR
  9  FROM CUSTOMERS C
 10  RIGHT JOIN ORDERS O
 11    ON C.CUST_ID = O.CUST_ID
 12  ORDER BY O.ORDER_DATE;

ORDER_REF ORDER_DATE ORDER_TOTAL ORDER_STATUS CUST_ID  CUSTOMER_NAME
--------- ---------- ----------- ------------ -------- --------------------
EMAIL_ADDR
------------------------------
ORD0001   10-JAN-26        55850 Delivered    CUST0001 Arjun Kumar
arjun.kumar@gmail.com

ORD0002   25-JAN-26         3350 Delivered    CUST0002 Priya Sharma
priya.sharma@gmail.com

ORD0003   05-FEB-26        28000 Delivered    CUST0003 Rahul Menon
rahul.menon@gmail.com

ORDER_REF ORDER_DATE ORDER_TOTAL ORDER_STATUS CUST_ID  CUSTOMER_NAME
--------- ---------- ----------- ------------ -------- --------------------
EMAIL_ADDR
------------------------------
ORD0004   18-FEB-26         8300 Delivered    CUST0004 Ananya Iyer
ananya.iyer@gmail.com

ORD0005   02-MAR-26        22000 Processing   CUST0005 Vikram Reddy
vikram.reddy@gmail.com

ORD0006   15-MAR-26         7000 Delivered    CUST0006 Meera Krishnan
meera.krishnan@gmail.com

ORDER_REF ORDER_DATE ORDER_TOTAL ORDER_STATUS CUST_ID  CUSTOMER_NAME
--------- ---------- ----------- ------------ -------- --------------------
EMAIL_ADDR
------------------------------
ORD0007   01-APR-26         6500 Shipped      CUST0007 Suresh Nair
suresh.nair@gmail.com

ORD0008   20-APR-26         5700 Delivered    CUST0008 Kavya Rajan
kavya.rajan@gmail.com

ORD0009   05-MAY-26        18500 Processing   CUST0009 Karthik Subramanian
karthik.subramanian@gmail.com

ORDER_REF ORDER_DATE ORDER_TOTAL ORDER_STATUS CUST_ID  CUSTOMER_NAME
--------- ---------- ----------- ------------ -------- --------------------
EMAIL_ADDR
------------------------------
ORD0010   18-MAY-26         5200 Delivered    CUST0010 Divya Pillai
divya.pillai@gmail.com

ORD0011   03-JUN-26         7000 Shipped      CUST0011 Aditya Verma
aditya.verma@gmail.com

ORD0012   20-JUN-26         2450 Delivered    CUST0012 Sneha Patel
sneha.patel@gmail.com

ORDER_REF ORDER_DATE ORDER_TOTAL ORDER_STATUS CUST_ID  CUSTOMER_NAME
--------- ---------- ----------- ------------ -------- --------------------
EMAIL_ADDR
------------------------------
ORD0014   15-AUG-26        28000 Delivered    CUST0014 Lakshmi Venkatesan
lakshmi.venkatesan@gmail.com

ORD0015   25-SEP-26         8350 Processing   CUST0015 Nithin Murugan
nithin.murugan@gmail.com

14 rows selected.
```

2. Find orders where customer details are missing.

```
SQL> SELECT
  2       O.ORDER_REF,
  3       O.ORDER_DATE,
  4       O.ORDER_TOTAL,
  5       O.ORDER_STATUS
  6  FROM CUSTOMERS C
  7  RIGHT JOIN ORDERS O
  8    ON C.CUST_ID = O.CUST_ID
  9  WHERE C.CUST_ID IS NULL;

no rows selected
```

3. Display all payment records with order details.

```
SQL> SELECT
  2       P.TXN_ID,
  3       P.TXN_DATE,
  4       P.PAY_METHOD,
  5       P.TXN_STATUS,
  6       P.AMOUNT_PAID,
  7       O.ORDER_REF,
  8       O.ORDER_DATE,
  9       O.ORDER_TOTAL,
 10       O.ORDER_STATUS
 11  FROM ORDERS O
 12  RIGHT JOIN ORDER_PAYMENTS P
 13    ON O.ORDER_REF = P.ORDER_REF
 14  ORDER BY P.TXN_DATE;

TXN_ID  TXN_DATE  PAY_METHOD      TXN_STATUS AMOUNT_PAID ORDER_REF ORDER_DATE ORDER_TOTAL ORDER_STATUS
------- --------- --------------- ---------- ----------- --------- ---------- ----------- ------------
TXN0001 10-JAN-26 Credit Card     Completed        55850 ORD0001   10-JAN-26        55850 Delivered
TXN0002 25-JAN-26 UPI             Completed         3350 ORD0002   25-JAN-26         3350 Delivered
TXN0003 18-FEB-26 Debit Card      Completed         8300 ORD0004   18-FEB-26         8300 Delivered
TXN0004 15-MAR-26 Net Banking     Completed         7000 ORD0006   15-MAR-26         7000 Delivered
TXN0005 20-APR-26 COD             Completed         5700 ORD0008   20-APR-26         5700 Delivered
TXN0006 18-MAY-26 UPI             Completed         5200 ORD0010   18-MAY-26         5200 Delivered
TXN0007 20-JUN-26 Credit Card     Completed         2450 ORD0012   20-JUN-26         2450 Delivered
TXN0009 14-AUG-26 Debit Card      Failed           28000 ORD0014   15-AUG-26        28000 Delivered
TXN0008 15-AUG-26 Debit Card      Completed        28000 ORD0014   15-AUG-26        28000 Delivered

9 rows selected.
```

5. Retrieve Complete Order Details

Task:

Generate a complete order report by combining:

- Customer Name
- Product Name
- Quantity
- Order Date
- Total Amount
- Payment Status

```
SQL> SELECT
  2       C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  3       P.ITEM_NAME,
  4       OD.ITEM_QTY,
  5       O.ORDER_DATE,
  6       O.ORDER_TOTAL,
  7       PM.TXN_STATUS
  8  FROM CUSTOMERS C
  9  JOIN ORDERS O
 10    ON C.CUST_ID = O.CUST_ID
 11  JOIN ORDER_ITEMS OD
 12    ON O.ORDER_REF = OD.ORDER_REF
 13  JOIN PRODUCTS P
 14    ON OD.PROD_ID = P.PROD_ID
 15  LEFT JOIN ORDER_PAYMENTS PM
 16    ON O.ORDER_REF = PM.ORDER_REF
 17  ORDER BY O.ORDER_DATE, C.CUST_ID;

CUSTOMER_NAME
--------------------------------------------------------------------------------
ITEM_NAME                           ITEM_QTY ORDER_DAT ORDER_TOTAL TXN_STATUS
----------------------------------- -------- --------- ----------- ----------
Arjun Kumar
Laptop                                     1 10-JAN-26       55850 Completed

Arjun Kumar
Wireless Mouse                             1 10-JAN-26       55850 Completed

CUSTOMER_NAME
--------------------------------------------------------------------------------
ITEM_NAME                           ITEM_QTY ORDER_DAT ORDER_TOTAL TXN_STATUS
----------------------------------- -------- --------- ----------- ----------
Priya Sharma
USB Cable                                  1 25-JAN-26        3350 Completed

Priya Sharma
Bluetooth Speaker                          1 25-JAN-26        3350 Completed

CUSTOMER_NAME
--------------------------------------------------------------------------------
ITEM_NAME                           ITEM_QTY ORDER_DAT ORDER_TOTAL TXN_STATUS
----------------------------------- -------- --------- ----------- ----------
Rahul Menon
Smartphone                                 6 05-FEB-26       28000

Ananya Iyer
USB Cable                                  4 18-FEB-26        8300 Completed

Ananya Iyer
Smart Watch                                1 18-FEB-26        8300 Completed
```

```
CUSTOMER_NAME
--------------------------------------------------------------------------------
ITEM_NAME                           ITEM_QTY ORDER_DAT ORDER_TOTAL TXN_STATUS
----------------------------------- -------- --------- ----------- ----------
Vikram Reddy
Tablet                                     1 02-MAR-26       22000

Meera Krishnan
Wireless Earbuds                           1 15-MAR-26        7000 Completed

Meera Krishnan
Headphones                                 1 15-MAR-26        7000 Completed

CUSTOMER_NAME
--------------------------------------------------------------------------------
ITEM_NAME                           ITEM_QTY ORDER_DAT ORDER_TOTAL TXN_STATUS
----------------------------------- -------- --------- ----------- ----------
Suresh Nair
Smart Watch                                1 01-APR-26        6500

Kavya Rajan
External Hard Disk                         1 20-APR-26        5700 Completed

Kavya Rajan
USB Cable                                  1 20-APR-26        5700 Completed

Karthik Subramanian
Gaming Monitor                             1 05-MAY-26       18500

CUSTOMER_NAME
--------------------------------------------------------------------------------
ITEM_NAME                           ITEM_QTY ORDER_DAT ORDER_TOTAL TXN_STATUS
----------------------------------- -------- --------- ----------- ----------
Divya Pillai
External Hard Disk                         1 18-MAY-26        5200 Completed

Aditya Verma
Wireless Earbuds                           1 03-JUN-26        7000

Aditya Verma
Headphones                                 1 03-JUN-26        7000

CUSTOMER_NAME
--------------------------------------------------------------------------------
ITEM_NAME                           ITEM_QTY ORDER_DAT ORDER_TOTAL TXN_STATUS
----------------------------------- -------- --------- ----------- ----------
Sneha Patel
Laptop Stand                               1 20-JUN-26        2450 Completed

Sneha Patel
Mechanical Keyboard                        1 20-JUN-26        2450 Completed

Nithin Murugan
Wireless Earbuds                           1 25-SEP-26        8350

20 rows selected.
```

6. Display Customer Purchase History

Tasks:

Write queries to find:

1. All products purchased by a customer.

```
SQL> SELECT
  2       C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  3       O.ORDER_REF,
  4       O.ORDER_DATE,
  5       P.ITEM_NAME,
  6       OD.ITEM_QTY,
  7       OD.UNIT_PRICE
  8  FROM CUSTOMERS C
  9  JOIN ORDERS O
 10    ON C.CUST_ID = O.CUST_ID
 11  JOIN ORDER_ITEMS OD
 12    ON O.ORDER_REF = OD.ORDER_REF
 13  JOIN PRODUCTS P
 14    ON OD.PROD_ID = P.PROD_ID
 15  WHERE C.CUST_ID = 'CUST0006'
 16  ORDER BY O.ORDER_DATE;

CUSTOMER_NAME                  ORDER_REF ORDER_DATE ITEM_NAME                    ITEM_QTY UNIT_PRICE
------------------------------ --------- ---------- ------------------------------ -------- ----------
Meera Krishnan                 ORD0006   15-MAR-26  Wireless Earbuds                    1       3500
Meera Krishnan                 ORD0006   15-MAR-26  Headphones                          1       4200
```

2. Total amount spent by each customer.

```
SQL> SELECT
  2       C.CUST_ID,
  3       C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4       ROUND(SUM(O.ORDER_TOTAL), 2) AS TOTAL_AMOUNT_SPENT
  5  FROM CUSTOMERS C
  6  JOIN ORDERS O
  7    ON C.CUST_ID = O.CUST_ID
  8  GROUP BY
  9       C.CUST_ID,
 10       C.FIRST_NAME,
 11       C.LAST_NAME
 12  ORDER BY TOTAL_AMOUNT_SPENT DESC;

CUST_ID  CUSTOMER_NAME                                    TOTAL_AMOUNT_SPENT
-------- ------------------------------------------------ ------------------
CUST0001 Arjun Kumar                                                   55850
CUST0014 Lakshmi Venkatesan                                            28000
CUST0003 Rahul Menon                                                   28000
CUST0005 Vikram Reddy                                                  22000
CUST0009 Karthik Subramanian                                           18500
CUST0015 Nithin Murugan                                                 8350
CUST0004 Ananya Iyer                                                    8300
CUST0006 Meera Krishnan                                                 7000
CUST0011 Aditya Verma                                                   7000
CUST0007 Suresh Nair                                                    6500
CUST0008 Kavya Rajan                                                    5700
CUST0010 Divya Pillai                                                   5200
CUST0002 Priya Sharma                                                   3350
CUST0012 Sneha Patel                                                    2450

14 rows selected.
```

3. Number of orders placed by each customer.

```
SQL> SELECT
  2       C.CUST_ID,
  3       C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4       COUNT(O.ORDER_REF) AS NUMBER_OF_ORDERS
  5  FROM CUSTOMERS C
  6  LEFT JOIN ORDERS O
  7    ON C.CUST_ID = O.CUST_ID
  8  GROUP BY
  9       C.CUST_ID,
 10       C.FIRST_NAME,
 11       C.LAST_NAME
 12  ORDER BY NUMBER_OF_ORDERS DESC;

CUST_ID  CUSTOMER_NAME                                        NUMBER_OF_ORDERS
-------- -------------------------------------------------- ----------------
CUST0001 Arjun Kumar                                                       1
CUST0015 Nithin Murugan                                                    1
CUST0003 Rahul Menon                                                       1
CUST0004 Ananya Iyer                                                       1
CUST0005 Vikram Reddy                                                      1
CUST0006 Meera Krishnan                                                    1
CUST0007 Suresh Nair                                                       1
CUST0008 Kavya Rajan                                                       1
CUST0009 Karthik Subramanian                                               1
CUST0010 Divya Pillai                                                      1
CUST0011 Aditya Verma                                                      1
CUST0012 Sneha Patel                                                       1
CUST0002 Priya Sharma                                                      1
CUST0014 Lakshmi Venkatesan                                                1
CUST0013 Rohan Bose                                                        0

15 rows selected.
```

4. Latest purchase details of customers.

```
SQL> SELECT
  2       C.CUST_ID,
  3       C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4       O.ORDER_REF,
  5       O.ORDER_DATE AS LATEST_PURCHASE_DATE,
  6       O.ORDER_TOTAL,
  7       O.ORDER_STATUS
  8  FROM CUSTOMERS C
  9  JOIN ORDERS O
 10    ON C.CUST_ID = O.CUST_ID
 11  WHERE O.ORDER_DATE = (
 12        SELECT MAX(O2.ORDER_DATE)
 13        FROM ORDERS O2
 14        WHERE O2.CUST_ID = C.CUST_ID
 15  )
 16  ORDER BY C.CUST_ID;

CUST_ID  CUSTOMER_NAME
--------------------------------------------------------------------------------
ORDER_REF LATEST_PU ORDER_TOTAL ORDER_STATUS
--------- --------- ----------- ------------
CUST0001 Arjun Kumar
ORD0001   10-JAN-26       55850 Delivered

CUST0002 Priya Sharma
ORD0002   25-JAN-26        3350 Delivered

CUST0003 Rahul Menon
ORD0003   05-FEB-26       28000 Delivered

CUST0004 Ananya Iyer
ORD0004   18-FEB-26        8300 Delivered

CUST0005 Vikram Reddy
ORD0005   02-MAR-26       22000 Processing

CUST0006 Meera Krishnan
ORD0006   15-MAR-26        7000 Delivered

CUST0007 Suresh Nair
ORD0007   01-APR-26        6500 Shipped

CUST0008 Kavya Rajan
ORD0008   20-APR-26        5700 Delivered

CUST0009 Karthik Subramanian
ORD0009   05-MAY-26       18500 Processing

CUST0010 Divya Pillai
ORD0010   18-MAY-26        5200 Delivered

CUST0011 Aditya Verma
ORD0011   03-JUN-26        7000 Shipped

CUST0012 Sneha Patel
ORD0012   20-JUN-26        2450 Delivered

CUST0014 Lakshmi Venkatesan
ORD0014   15-AUG-26       28000 Delivered

CUST0015 Nithin Murugan
ORD0015   25-SEP-26        8350 Processing

14 rows selected.
```

7. Generate Multi-Table Business Reports

Report 1: Customer Order Report

Display:

- Customer name
- Order ID
- Order date
- Order status

```
SQL> SELECT
  2       C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  3       O.ORDER_REF,
  4       O.ORDER_DATE,
  5       O.ORDER_STATUS
  6  FROM CUSTOMERS C
  7  JOIN ORDERS O
  8    ON C.CUST_ID = O.CUST_ID
  9  ORDER BY O.ORDER_DATE;

CUSTOMER_NAME                  ORDER_REF ORDER_DATE ORDER_STATUS
------------------------------ --------- ---------- ------------
Arjun Kumar                    ORD0001   10-JAN-26  Delivered
Priya Sharma                   ORD0002   25-JAN-26  Delivered
Rahul Menon                    ORD0003   05-FEB-26  Delivered
Ananya Iyer                    ORD0004   18-FEB-26  Delivered
Vikram Reddy                   ORD0005   02-MAR-26  Processing
Meera Krishnan                 ORD0006   15-MAR-26  Delivered
Suresh Nair                    ORD0007   01-APR-26  Shipped
Kavya Rajan                    ORD0008   20-APR-26  Delivered
Karthik Subramanian            ORD0009   05-MAY-26  Processing
Divya Pillai                   ORD0010   18-MAY-26  Delivered
Aditya Verma                   ORD0011   03-JUN-26  Shipped
Sneha Patel                    ORD0012   20-JUN-26  Delivered
Lakshmi Venkatesan             ORD0014   15-AUG-26  Delivered
Nithin Murugan                 ORD0015   25-SEP-26  Processing

14 rows selected.
```

Report 2: Sales Report

Display:

- Product name
- Quantity sold
- Total revenue

```
SQL> SELECT
  2       P.ITEM_NAME,
  3       SUM(OD.ITEM_QTY) AS QUANTITY_SOLD,
  4       SUM(OD.ITEM_QTY * OD.UNIT_PRICE) AS TOTAL_REVENUE
  5  FROM PRODUCTS P
  6  JOIN ORDER_ITEMS OD
  7    ON P.PROD_ID = OD.PROD_ID
  8  GROUP BY P.PROD_ID, P.ITEM_NAME
  9  ORDER BY TOTAL_REVENUE DESC;

ITEM_NAME                                          QUANTITY_SOLD TOTAL_REVENUE
-------------------------------------------------- ------------- -------------
Smartphone                                                     6        168000
Laptop                                                         1         55000
Tablet                                                         1         22000
Gaming Monitor                                                 1         18500
Smart Watch                                                    2         13000
Wireless Earbuds                                               3         10500
External Hard Disk                                             2         10400
Headphones                                                     2          8400
Bluetooth Speaker                                              1          3200
USB Cable                                                      6          2700
Mechanical Keyboard                                            1          2500
Laptop Stand                                                   1          1500
Wireless Mouse                                                 1           850

13 rows selected.
```

Report 3: Payment Analysis Report

Display:

- Payment mode
- Number of transactions
- Successful payments

```
SQL> SELECT
  2       PAY_METHOD,
  3       COUNT(*) AS NUMBER_OF_TRANSACTIONS,
  4       SUM(
  5            CASE
  6                 WHEN TXN_STATUS = 'Completed' THEN 1
  7                 ELSE 0
  8            END
  9       ) AS SUCCESSFUL_PAYMENTS
 10  FROM ORDER_PAYMENTS
 11  GROUP BY PAY_METHOD
 12  ORDER BY NUMBER_OF_TRANSACTIONS DESC;

PAY_METHOD         NUMBER_OF_TRANSACTIONS SUCCESSFUL_PAYMENTS
------------------ ---------------------- -------------------
Debit Card                              3                   2
UPI                                     2                   2
Credit Card                             2                   2
COD                                     1                   1
Net Banking                             1                   1
```

Report 4: Customer Purchase Analysis

Find:

- Top purchasing customers

```
SQL> SELECT
  2       C.CUST_ID,
  3       C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4       SUM(O.ORDER_TOTAL) AS TOTAL_SPENDING
  5  FROM CUSTOMERS C
  6  JOIN ORDERS O
  7    ON C.CUST_ID = O.CUST_ID
  8  GROUP BY
  9       C.CUST_ID,
 10       C.FIRST_NAME,
 11       C.LAST_NAME
 12  ORDER BY TOTAL_SPENDING DESC;

CUST_ID  CUSTOMER_NAME                                    TOTAL_SPENDING
-------- ------------------------------------------------ --------------
CUST0001 Arjun Kumar                                             55850
CUST0014 Lakshmi Venkatesan                                      28000
CUST0003 Rahul Menon                                             28000
CUST0005 Vikram Reddy                                            22000
CUST0009 Karthik Subramanian                                     18500
CUST0015 Nithin Murugan                                           8350
CUST0004 Ananya Iyer                                              8300
CUST0006 Meera Krishnan                                           7000
CUST0011 Aditya Verma                                             7000
CUST0007 Suresh Nair                                              6500
CUST0008 Kavya Rajan                                              5700
CUST0010 Divya Pillai                                             5200
CUST0002 Priya Sharma                                             3350
CUST0012 Sneha Patel                                              2450

14 rows selected.
```

- Customers with maximum orders

```
SQL> SELECT
  2       C.CUST_ID,
  3       C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4       COUNT(O.ORDER_REF) AS NUMBER_OF_ORDERS
  5  FROM CUSTOMERS C
  6  JOIN ORDERS O
  7    ON C.CUST_ID = O.CUST_ID
  8  GROUP BY
  9       C.CUST_ID,
 10       C.FIRST_NAME,
 11       C.LAST_NAME
 12  HAVING COUNT(O.ORDER_REF) = (
 13       SELECT MAX(ORDER_COUNT)
 14       FROM (
 15            SELECT COUNT(*) AS ORDER_COUNT
 16            FROM ORDERS
 17            GROUP BY CUST_ID
 18       )
 19  );

CUST_ID  CUSTOMER_NAME                                      NUMBER_OF_ORDERS
-------- -------------------------------------------------- ----------------
CUST0001 Arjun Kumar                                                       1
CUST0002 Priya Sharma                                                      1
CUST0003 Rahul Menon                                                       1
CUST0004 Ananya Iyer                                                       1
CUST0005 Vikram Reddy                                                      1
CUST0006 Meera Krishnan                                                    1
CUST0007 Suresh Nair                                                       1
CUST0008 Kavya Rajan                                                       1
CUST0009 Karthik Subramanian                                               1
CUST0010 Divya Pillai                                                      1
CUST0011 Aditya Verma                                                      1
CUST0012 Sneha Patel                                                       1
CUST0014 Lakshmi Venkatesan                                                1
CUST0015 Nithin Murugan                                                    1

14 rows selected.
```

- Customers with highest spending

```
SQL> SELECT
  2       C.CUST_ID,
  3       C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4       SUM(O.ORDER_TOTAL) AS TOTAL_SPENDING
  5  FROM CUSTOMERS C
  6  JOIN ORDERS O
  7    ON C.CUST_ID = O.CUST_ID
  8  GROUP BY
  9       C.CUST_ID,
 10       C.FIRST_NAME,
 11       C.LAST_NAME
 12  HAVING SUM(O.ORDER_TOTAL) = (
 13       SELECT MAX(TOTAL_SPENDING)
 14       FROM (
 15            SELECT SUM(ORDER_TOTAL) AS TOTAL_SPENDING
 16            FROM ORDERS
 17            GROUP BY CUST_ID
 18       )
 19  );

CUST_ID  CUSTOMER_NAME                                    TOTAL_SPENDING
-------- -------------------------------------------------- --------------
CUST0001 Arjun Kumar                                             55850
```